Sahil Tushar Hate

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Education

University of Southern California August 2024 - May 2026
Master of Science in Computer Science *Los Angeles, California*
Relevant Coursework: Algorithms, Database Systems, Natural Language Processing, Machine Learning, Information Retrieval

University of Mumbai August 2019 - May 2023
Bachelor of Technology in Information Technology *Mumbai, India*
Relevant Coursework: Data Structures, Algorithms, Operating Systems, Cloud Computing, Information & Network Security

Technical Skills

Languages: Java, Python, C/C++, Go, JavaScript, SQL
Frameworks & Technologies: Node.js, FastAPI, React.js, Django, REST APIs, gRPC, WebSockets, Kafka, TCP/IP
Databases: PostgreSQL, MongoDB, Redis, ChromaDB
Cloud and Infrastructure: Google Cloud Platform (GCP), Oracle Cloud, AWS, Docker, Kubernetes
Machine Learning / AI: TensorFlow, OpenCV, PyTorch, Keras, LangChain, RAG, LLMs, Agentic AI
Developer Tools & Testing: Git, GitHub Actions, Jenkins, Linux, Bash/Shell Scripting, JUnit

Experience

Medtronic (now MiniMed) February 2026 - May 2026
Software Development Engineer Co-Op *Los Angeles, California*

- Developed a **CGM sensor simulation application** supporting device pairing and real-time event simulation, and collaborated with cross-functional teams to resolve software issues through log analysis, execution tracing, and protocol validation.
- Designed **end-to-end validation and regression test scenarios** across **10 engineering workflows**, improving software reliability and ensuring robust behavior for safety-critical diabetes management features.

Qualcomm May 2025 - August 2025
Software Engineer Intern *San Diego, California*

- Designed and deployed a **distributed multi-agent AI platform** using Python, LangGraph, and Claude Opus 4, implementing modular execution pipelines, contextual memory management, and dynamic task orchestration across **8 specialized agents**, improving complex query resolution accuracy by **65%**.
- Built a **scalable retrieval infrastructure** capable of indexing and serving **100K+ vector embeddings** across **250+ enterprise data sources** using FAISS and ChromaDB, reducing semantic search latency from ~7s to **under 2s**.
- **Optimized production APIs** through request pipeline profiling, asynchronous execution, and improved resource utilization, reducing inference latency by **20%** while supporting thousands of concurrent requests.

Oracle June 2023 - July 2024
Software Engineer *Mumbai, India*

- Architected **scalable Java EE-based microservices** for Oracle Banking Digital Experience (OBDX), supporting **core banking modules** and high-volume transactions for financial institutions.
- Engineered secure authentication services by integrating **multi-factor authentication (MFA) and adaptive risk-based access control** into enterprise financial systems, strengthening platform security while improving transactional performance.
- Collaborated with **cross-functional** teams to **redesign API architectures**, reducing integration latency by **~500 ms**.
- Streamlined **CI/CD operations using Jenkins** for automated build and deployment of Java EE applications on **Oracle Cloud Infrastructure (OCI)**, while leading root-cause analysis of production incidents that improved platform uptime by 20%.

Projects

Distributed Data Analytics Engine | *Python, FastAPI, Redis, PostgreSQL, Docker, React*

- Architected a scalable data processing system that ingested **100K+ row datasets**, executing asynchronous analysis via a **Redis-backed task queue** and scalable worker pool, achieving **~3×** latency reduction through parallel processing.
- Designed a **MapReduce-style aggregation pipeline** to merge partial metrics into real-time global analytics.
- Built real-time job monitoring by streaming processing metrics to a React dashboard using **Redis Pub/Sub and WebSockets**.

Real-Time Messaging System | *Node, React, socket.io, WebSockets, Message Queues*

- Engineered a **horizontally scalable** real-time messaging backend in Node.js and TypeScript, leveraging WebSockets and socket.io**,** sustaining sub-500 ms latency at 1,000+ messages/hour.
- Built **Redis-backed socket.io adapters** for queuing, ordering, and state synchronization across distributed Node.js instances.

Digital Carbon Footprint Platform | *Java, Python, Flask, REST APIs, PostgreSQL*

- Built a **modular prediction pipeline** integrating feature engineering and machine learning inference services, achieving **86% prediction accuracy** while supporting concurrent API requests through stateless backend design.

Publications

- “Implementation of Digital Carbon Footprint Emission Estimator and Predictor for Mobile Devices,” 6th **IEEE** - International Conference on Advances in Science & Technology (IEEE-ICAST 2023). Published in IEEE Xplore Digital Library.